

Sprint Report 1

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Intended Progress:

The primary goal of Sprint 1 was to establish the development environment, become familiar with the existing Spherical Easel codebase, and begin experimenting with possible approaches for implementing tours. Because the team is still becoming familiar with Vue, TypeScript, and the existing application, this sprint was intended to focus on exploration and prototyping rather than building the final tour system.

Wyatt: My plan for this sprint was to explore the existing GitHub codebase, help establish functioning local development environments for the team, and create an initial tour prototype. The prototype was intended to give the team something functional to work with while also helping us better understand Vue, the existing application structure, and how a tour system could interact with Spherical Easel.

Sam: My plan for this first sprint was to familiarize myself with the existing repository provided by Dr. Dulimarta and Dr. Dickinson, as well as to familiarize myself with Vue and Typescript/Javascript. I also wanted to ensure that I had set up both my laptop and my desktop to be able to edit on both by installing the necessary packages, such as Bun and Node.js.

Ben: My plan for Sprint 1 was to get the project running on my computer, get familiar with the existing Spherical Easel code, and build a small tour prototype from the existing UITour.vue component.

Jason: My plan for Sprint 1 was to understand the existing code and learn the basics of Vue and TypeScript because this is my first time doing anything with UI. I also wanted to get the program to run on my laptop.

Progress Reflection:

The team successfully established functioning local development environments and was able to clone and run the project locally. Getting the application's dependencies configured and determining the correct commands for running the tutorial demo took some time and required assistance from Dr. Dulimarta. Once the environment was working, the team was able to begin experimenting with the existing application.

Wyatt: The team was able to successfully clone and run Spherical Easel locally. After resolving the initial dependency and setup issues, working with my own fork of the repository went well. I also spent time determining how `UITour.vue` could function within the existing codebase and how a user might interact with tours.

One challenge was determining how the tour system should fit into the existing application without having a complete understanding of the codebase yet. Because of this, the prototype was intentionally focused on functionality rather than polish or final design. The primary goal was to produce something working that could be demonstrated to the sponsors and used to inform our future development.

My previous experience with Vue allowed me to develop the initial prototype relatively quickly. The prototype gave me a better understanding of how a tour could interact with the application's UI and provided the team with a concrete example to discuss. More importantly, the sprint showed that we need to spend additional time learning the existing codebase and Vue before committing to a final architecture for the tour system.

Sam: I'm still trying to get the hang of working with Vue in general. So far, I've understood the general structure of what a Vue file is like (HTML, Script, CSS), and the essentials of each section. I've also gone through the VTooltip documentation to see if it is a viable option for our tutorials, and decided that it may not be the best option (for now). I've continued to try and get my footing on Vue by watching many different YouTube tutorials, as well as other documentation.

Ben: First was getting the app set up to be able to test locally. Everything went pretty smoothly with installing Node.js and Bun in order for the environment to be set up. Then I spent a while getting familiar with the source code and reading through how an existing tour works in UITour.vue. And how the /tutorial link turns it on. I watched quite a bit of YouTube videos and just read things online to familiarize myself with the concepts and languages that are included in this project. Overall it's a step in the right direction as I am more confident I can dive into implementing more tours now that the environment is set up.

Jason: I spent most of my time getting familiar with Vue by reading documentation and watching YouTube tutorials. I also got the Spherical Easel to run locally on my laptop. I spent some time looking through the rest of the code and thinking of ideas for how I could check for user input. I didn't get to making any prototypes for tours; I'm planning on working on this later this week.

Problems Encountered:

The largest challenge during this sprint was the team's varying familiarity with Vue and the existing Spherical Easel codebase. While the development environment was successfully established, several team members expressed that they still need additional time to become comfortable developing with Vue before they can work efficiently on the tour system.

Wyatt: I underestimated how much time some team members would need to become comfortable with Vue and the existing codebase. My initial approach was to have each team member independently create a small prototype so that everyone could explore different approaches at their own pace. While this provided individual experience and produced useful examples, it did not provide as much collaborative learning as I initially expected.

If I were to repeat this sprint, I would structure the prototype work more collaboratively. For example, the team could have been divided to develop one or two prototypes together, allowing team members with different levels of Vue experience to work through problems together. This would likely have provided more opportunities for shared learning and reduced the amount of time some members spent trying to understand the framework independently.

For the next sprint, I will account for this by providing additional Vue resources and encouraging more collaboration when developing the tours. I will also make sure that tasks are assigned with the team's current familiarity with Vue in mind rather than assuming that everyone can immediately work at the same development pace.

Sam: I didn't think I would have this much trouble trying to learn and understand Vue. I thought having only JavaScript knowledge would be sufficient. That, unfortunately, isn't the case. There is much more to it than I thought, and it is something that I want to better understand and get a stronger foundation on. I underestimated how much time I would need to be able to learn and understand Vue. I need to ensure that I give ample time to this project, because underestimating things can be detrimental to the project. I want to go over with the team (who may have more Vue knowledge than me) and see if they can guide me more over how it works.

Ben: While I do have experience with working on websites and using Vue, it's been a while. So diving back into these things with the scope of a large project was a little overwhelming. But things are starting to come back to me as well as I am learning new things of course. The setup as I said did go smoothly but the site itself took way longer to load than I expected. And the tour I made, while very simple, has some issues with it that I'll try to understand and resolve in the coming days.

Jason: I don't have any experience with JavaScript, TypeScript, or Vue, so trying to learn all of them and then using them to add to existing code is a lot to handle in the first few weeks. I thought I could learn them fairly easily in this short amount of time, but it is taking me longer than I initially planned to understand it.

Projected Progress:

The team has established the basic development environment and has begun experimenting with tour implementations. The primary adjustment for the next sprint is to allow more time for learning Vue and the existing codebase while moving from individual prototypes toward a more consistent approach for developing tours.

Wyatt: For the next sprint, I plan to focus on helping the team move from experimentation toward a consistent method for creating tours. This includes providing useful Vue resources, helping team members understand the existing codebase, and using the prototypes from this sprint to identify common functionality that can eventually become part of a reusable tour framework.

The next sprint should therefore focus less on creating unrelated individual prototypes and more on comparing what we have learned, establishing a basic structure for tours, and beginning to develop reusable components. Our velocity will continue to be evaluated as the team gains more experience with the codebase and Vue.

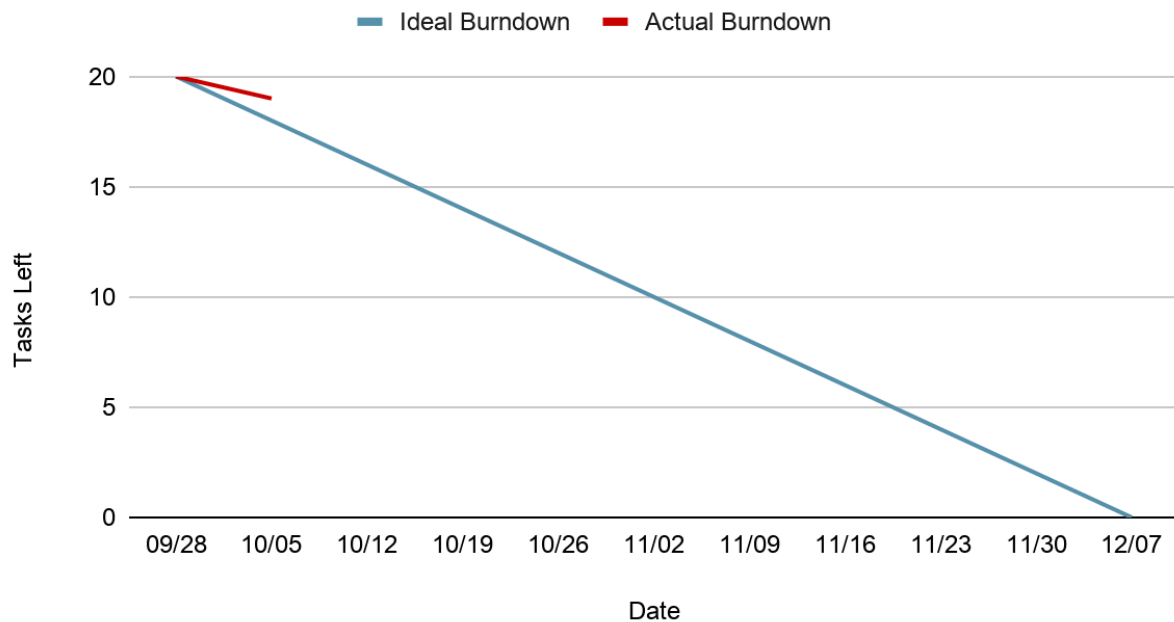
Sam: The main progress that's been done has been establishing our environments and (trying) to understand the codebase of the existing Spherical Easel app. I have been unable to create a prototype yet for a tutorial on the website. I believe that while being only 1 task behind, we are still on good pacing with the project, since these first few tasks of setup shouldn't be as difficult to fall behind on.

Ben: Next sprint will be focused on cleaning up this tour I made and simply working with the codebase more to better understand how to make bigger and better tours. Along with this, making a crystalized plan with our team to try and decide how these tours will be created will also be important to try and tackle. Because while this project is going very well so far, the structure isn't exactly set in stone yet.

Jason: For the next sprint, I am planning on meeting with the team and coming up with a structure for how the tours will be created. Once we come up with an established structure it should we be able to do a lot of tasks quickly. I plan on creating a specific tours from the list of tours that we were given by Dr. Dulimarta and Dr. Dickinson.

Update of burn down chart:

Ideal Burndown and Actual Burndown



Teamwork Reflections:

Wyatt: One area the team needs to improve is communicating task requirements and questions earlier in the sprint. Tasks should be assigned with enough lead time for team members to ask questions, become familiar with the relevant part of the codebase, and complete the work without having to rush near the end of the sprint.

We also need to increase communication when team members encounter difficulties with Vue, the existing codebase, or the project requirements. Asking questions earlier will make it easier for the rest of the team to help and will reduce the amount of time individuals spend blocked on a problem.

Going forward, I will focus on communicating upcoming tasks earlier, making sure responsibilities are clearly assigned, and encouraging team members to raise questions or problems as soon as they encounter them.

Sam: I think an area the team can improve on is communication. I know that I don't communicate as much as I should (especially if I need help with learning Vue), and I know the team as a whole can benefit greatly from more communication. I also believe that we may need better structure going forward on what we should expect from one another, and maybe having end of week check-ins before we do our weekly meetings on Tuesdays with our sponsors. I need to get better as well with time management, and not putting all of my work off until the end of sprints.

Ben: As said by most people who work in group projects including the members in mine, communication is extremely important when it comes to how smoothly a project can run. Without everyone being on the same page at least somewhat, disasters can sometimes happen. And myself included, I need to vastly improve my communication on this project with my team, because it hasn't exactly been good so far. Things are getting done, but we shouldn't have to worry about what people are working on and if it'll get done or not. And while I'm not the one to just not work on something, I shouldn't wait until the last minute either and cause any unneeded stress to the rest of my team. Having more check-ins would be a great first step.

Jason: I need to improve on communicating to the team what I am working on and what questions I have about the code. I spent a lot of the sprint learning, and I probably could have been more efficient if I asked the team questions about the things that confused me. I also need to improve my time management skills. I procrastinated a lot of my work and waited until a day or two before our weekly meetings to work on what I needed to. This led to unnecessary stress and caused me to rush.

Conclusion: Brief overview of the status of the group and project overall.

The team has established the development environment and has begun producing functional tour prototypes. Although the team did not complete as many tours as originally intended, the sprint provided useful experience with the existing codebase, Vue, and the potential structure of the tour system.

Wyatt: Overall, I believe the team is in a good position to begin developing the tours. The primary remaining challenge is building enough familiarity with Vue and the existing Spherical Easel codebase for each team member to work effectively. The next sprint should focus on addressing that learning curve while using what we learned from the initial prototypes to establish a more consistent and reusable approach to tour development.

Sam: I believe that the team is in good status at the moment. We are still early on in our development and I feel confident that we can work together to choose a prototype that we can all work together on. Having one prototype to work on will make communication easier within the team so that we can all work together instead of individually (which is something I was struggling with).

Ben: I think we're in a good spot right now. Things need to be improved a little bit in certain areas, but it could be so much worse. We all have our environments up and running which is great, and now we're diving head first into the main development of the project now that a lot of the logistics are done. My prototype right now for the point tour is very small, but it shows the idea of a basic tour and I'm interested to see where we go from here in the coming weeks.

Jason: I think we're off to a good start. I am more confident with Vue. As I continue to work on the project, I will get better at Vue and be able to finish tasks faster. It helps other team members who have prior experience with Vue, so we already have a prototype that I can reference and learn from. This next sprint will require more collaboration as we come up with a framework for the tours.